Foundation II: Shared Manifestation
A Conditional Two-Source Program for Shared Accessible Sectors

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Abstract

Foundation I of the Consciousness Series [1] constructs a conditional chain from a single source datum $S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\})$ to an admissible accessible pair $(\mathcal{A}_c, \omega)$, which then feeds into the inseparability theorem of Paper I [2]. That chain is single-source: it produces one observer's manifest sector. The present paper, Foundation II, asks the next structural question: when two source chains share the same ambient substrate, under what conditions does a non-trivial *shared* manifest sector emerge?

We define a multi-source datum as a family of source data sharing a common ambient triple $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$, each equipped with its own decoherence semigroup. No global tensor-product decomposition is assumed; the multi-source structure is encoded entirely through subalgebra inclusions, following the non-canonicality of tensor-product factorizations in the Type III_1 setting established in [5]. We then introduce joint-stability conditions under which a candidate shared sector $\mathcal{A}_{\text{sh}}$ is simultaneously stable for both sources.

The paper's main result is a conditional two-source foothold, not a general multi-source closure theorem. Under named joint-stability and compatible-projection conditions, we construct a shared stable hull $\Phi^{\text{st}}_{\text{sh}}(\mathfrak{F}_{\text{sh}})$ from the overlap of the two foreground frames, prove that it is a von Neumann subalgebra carrying a faithful restricted state, and show that, under an additional admissibility input, the resulting pair $(\Phi^{\text{st}}_{\text{sh}}(\mathfrak{F}_{\text{sh}}), \omega^{\text{st}}_{\text{sh}})$ lies in the theorem domain of Paper I. The shared sector therefore inherits the geometry–modular-flow inseparability of the single-source case.

After the main theorem, we analyze a regime-dependent common/private structure: under a shared/private regime assumption, the shared sector is the locus on the overlap algebra where the two manifest states agree, while observer-dependent discrepancy can only appear outside that shared sector. Finally, we state a conditional N-source extension under pairwise-threshold and exchangeability assumptions, and record the open problems that remain, including the quantitative relationship between consensus strength and geometric stability.

This paper treats only the mathematical bridge from multi-source data to shared accessible sectors. It does not claim a general N-source theorem, does not provide a quantitative consensus–stability bound, and does not derive the candidate shared sector from bare multi-source dynamics alone. It also does not attempt to settle questions of intersubjective ontology.

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1 Introduction

Foundation I of the Consciousness Series [1] addressed the single-source problem. Starting from a source datum

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

it isolated a conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where $\mathcal{O}_A$ is a persistent projected record process, $\mathcal{O}_B$ is a foreground frame supplied under explicit finite-coordinate coding assumptions, and $(\mathcal{A}_c, \omega)$ is the manifest accessible pair that then feeds into the geometry–modular-flow inseparability theorem of Paper I [2]. That was the correct scope for the first foundation paper: it explained how one observer-facing manifest sector could arise from one source chain.

The present paper asks the next structural question. If two source chains share the same ambient substrate, under what conditions does a non-trivial *shared* manifest sector arise? This is the minimum multi-source problem behind any later discussion of a shared world. The paper does not attempt to settle the full intersubjective problem in one step. It does not claim a general N-source theorem, does not derive a quantitative consensus-strength/stability law, and does not aim to settle questions of ontology. Its narrower aim is to identify a theorem-level foothold for *shared manifestation existence*.

That calibration matters. In the current programme there are two separate bridges beyond the already-written Consciousness Trilogy results. The first concerns whether entangled or correlated source chains can support a common manifest sector at all. The second concerns how the strength of that commonality controls the quantitative stability of the resulting geometry. Foundation II addresses only the first. The second remains a separate Bridge 3 problem and should not be smuggled into the present paper under vague language about "consensus" or "robustness".

The paper therefore adopts the smallest honest target. Its main theorem is proved in the two-source case. The theorem does not directly state that a full shared geometry has been derived. Its first conclusion is weaker and mathematically safer: under named compatibility and joint-stability inputs, one can construct a *shared admissible manifest sector*

$$(\mathcal{A}_{\text{sh}}, \omega_{\text{sh}}) \subseteq (\mathcal{A}_c^{(1)}, \omega_1) \cap (\mathcal{A}_c^{(2)}, \omega_2),$$

which is stable under the relevant source-side dynamics and can then be fed into the already-existing Paper I machinery. Geometry enters only as a corollary once that admissible shared sector is in hand. The main result should therefore be read as a conditional inclusion theorem and theorem-level foothold, not as a full emergence theorem for shared worlds.

At the heuristic level, this shared-sector question is related to the quantum Darwinism and spectrum-broadcast-structure literature, where the objective part of a quantum description is identified with redundantly accessible pointer information carried across multiple environment fragments [17, 18, 19, 20]. The present paper does not translate those frameworks into a theorem about Type III_1 ambient algebras. It only records the operator-algebraic analogue needed here: a shared foreground overlap together with a shared stable hull.

Three design choices govern the whole paper.

1. **Two sources first.** The paper starts at $N = 2$, because the two-source case is the minimum non-trivial setting in which one can ask whether a shared manifest sector exists at all. General N-source statements are postponed to a conditional extension section.

2. **No tensor-product starting point.** The multi-source datum is formulated inside a common ambient algebra rather than from a canonical tensor decomposition. This follows the HAFF viewpoint that physically relevant subsystem structure is selected by stability and accessibility rather than fixed in advance [5, 6].

3. **Common/private decomposition is secondary.** After the main theorem, we analyze how each observer's full manifest algebra may be written, under a regime assumption, as a shared sector plus a private complement. That decomposition is treated as a structural consequence, not as a competing main theorem.

The paper's mathematical contributions are correspondingly limited and explicit.

1. We define a *multi-source datum* on a common standard-form von Neumann algebra, with Type III_1 as the primary regime, together with independent faithful normal source states and source-specific decoherence semigroups.

2. We separate the new inputs into three named joint-stability conditions: shared-sector invariance, shared-state existence with faithfulness, and projection/kernel compatibility.

3. In the two-source case, we define the shared foreground overlap and the associated shared stable hull, and prove the abstract minimality properties of that hull.

4. Under an explicit candidate shared-sector input, we prove an inclusion theorem showing that the shared stable hull lies inside the common manifest carrier, and therefore yields a shared admissible manifest sector.

5. Under a shared/private regime assumption, we analyze the resulting common/private structure by identifying the shared sector as the locus of agreement on the overlap algebra and by localizing overlap-level discrepancy outside that sector.

The role of conditional language is exactly the same as in Foundation I. Wherever a piece can be internalized cleanly, we internalize it. Wherever it cannot, we isolate it and keep it explicit. The present theorem does not derive the existence of a candidate shared sector from bare multi-source dynamics or entanglement alone. It starts from such a sector as named input and analyzes the stable closure that follows from it. The main open points here are not hidden: model-independent joint stability, compatible Nakajima–Zwanzig projections, and the well-definedness of common/private decomposition in the Type III_1 setting.

The paper is organized as follows. Section 2 defines the multi-source datum and the joint-stability package. Section 3 states and proves the two-source main theorem. Section 4 analyzes the common/private structure under a regime assumption. Section 5 gives a conditional N-source extension. Section 6 records the open problems. Section 7 collects the named conditional inputs. Appendix A summarizes the imported Genesis and Consciousness results used by the paper.

2 Multi-Source Datum and Joint-Stability Inputs

The first task is to define the source-side object language without overloading it with later conditional interface data.

Definition 2.1 (Multi-source datum)**.** A *multi-source datum* is a tuple

$$S_{\text{multi}} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_i\}_{i=1}^N, \{\mathcal{E}_t^{(i)}\}_{i=1}^N),$$

where:

1. $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a von Neumann algebra in standard form, with Ω cyclic and separating for $\mathcal{A}_U$; the programme's primary infinite-dimensional regime is a Type III_1 factor, while explicitly marked model checks may be finite Type I;

2. $\varpi_1, \ldots, \varpi_N$ are independently specified faithful normal states on $\mathcal{A}_U$; no agreement of their restrictions to later overlap algebras is part of the datum;

3. for each $i \in \{1, \ldots, N\}$, $\{\mathcal{E}_t^{(i)}\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$.

Remark 2.2 (Heisenberg-picture convention). The semigroups $\{\mathcal{E}_t^{(i)}\}$ act on observables, so the paper uses the Heisenberg-picture convention throughout. The continuity assumption in Definition 2.1 is therefore pointwise σ-weak continuity on $\mathcal{A}_U$, not norm continuity.

Remark 2.3 (A deliberately thin datum). The multi-source datum contains only the common ambient standard-form triple, the independent source states, and the family of source-side dynamical semigroups. It does *not* include Nakajima–Zwanzig projections, foreground frames, accessible algebras, or any private/shared decomposition. Those are not primitive data; they are interface or output objects that may or may not be obtainable from the datum under additional assumptions.

Remark 2.4 (Why no tensor-product factorization). The present paper does not assume a global tensor-product decomposition

$$\mathcal{H}_U \cong \bigotimes_{i=1}^N \mathcal{H}_i \qquad \text{or} \qquad \mathcal{A}_U \cong \bar{\otimes}_{i=1}^N \mathcal{A}_i.$$

This is deliberate. In the HAFF programme, the physically relevant subalgebras are selected by accessibility and stability rather than by a canonical prior factorization [5, 6]. In the Type III setting, treating the subsystem split as primitive would import exactly the kind of structure the Genesis programme is designed to derive rather than presuppose.

The single-source outputs from Foundation I are now imported only as downstream conditional objects.

Definition 2.5 (Individual manifest outputs)**.** Let S_{multi} be a multi-source datum. For each source index i, an *individual manifest output* means a pair

$$(\mathcal{A}_c^{(i)}, \omega_i)$$

obtained from the Foundation I chain for the i-th source, where $\mathcal{A}_c^{(i)} \subseteq \mathcal{A}_U$ is a von Neumann subalgebra and

$$\omega_i := \varpi_i|_{\mathcal{A}_c^{(i)}}$$

is its faithful normal manifest state.

Remark 2.6 (Independent states are load-bearing). The states ω_i are not all restrictions of the reference vector state $\langle \Omega, \cdot\, \Omega \rangle$. The vector Ω fixes a standard-form representation and the reference modular flow used below; the ϖ_i carry source-specific state data. If instead one defined every ω_i by restricting a single ambient state, then the states would agree automatically on every intersection and the overlap discrepancy of Section 4 would vanish identically.

Remark 2.7. Nothing in Definition 2.3 claims that every source in a multi-source datum automatically produces an individual manifest output. That is already the conditional content of Foundation I [1]. Foundation II takes those single-source outputs as antecedent material when the two-source main theorem is stated.

Definition 2.8 (Candidate shared sector)**.** Given individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$, a *candidate shared sector* is a von Neumann subalgebra

$$\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \bigcap_{i=1}^{N} \mathcal{A}_c^{(i)}.$$

The new source-side content begins with the joint-stability package. We state it in three layers so that later failures can be localized cleanly.

Definition 2.9 (JS-A: shared-sector invariance and semigroup compatibility)**.** Let S_{multi} be a multi-source datum and let $\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \mathcal{A}_U$ be a candidate shared sector. We say that $(\mathcal{A}_{\text{sh}}^{\text{cand}}, \{\mathcal{E}_t^{(i)}\})$ satisfies *JS-A* if:

1. $\mathcal{E}_t^{(i)}(A) \in \mathcal{A}_{\text{sh}}^{\text{cand}}$ for every $A \in \mathcal{A}_{\text{sh}}^{\text{cand}}$, every i, and every $t \geq 0$;
2. the ambient modular flow $\sigma_s^{\omega_0,\text{tot}}$ of the reference state $\omega_0(\cdot) = \langle \Omega, \cdot\, \Omega \rangle$ satisfies
$$\sigma_s^{\omega_0,\text{tot}}(\mathcal{A}_{\text{sh}}^{\text{cand}}) = \mathcal{A}_{\text{sh}}^{\text{cand}} \qquad \forall s \in \mathbb{R};$$
3. the restricted semigroups commute on the candidate shared sector:
$$\mathcal{E}_t^{(i)} \mathcal{E}_u^{(j)}(A) = \mathcal{E}_u^{(j)} \mathcal{E}_t^{(i)}(A) \qquad \forall A \in \mathcal{A}_{\text{sh}}^{\text{cand}},\ \forall t, u \geq 0.$$

Remark 2.10 (Status of the JS-A commutativity clause). The exact commutativity requirement in Definition 2.9 is a strong regime assumption. The present paper does not claim that it follows model-independently from the bare multi-source datum. A weaker variant would replace strict commutativity by asymptotic commutativity in the long-time limit, or by commutativity only on the shared fixed-point algebra. The stronger form is used here because it makes the shared stable-carrier construction transparent; the weaker variants are left open.

Definition 2.11 (JS-B: shared-state existence and faithfulness)**.** Let $\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \mathcal{A}_U$ be a candidate shared sector. We say that $\mathcal{A}_{\text{sh}}^{\text{cand}}$ satisfies *JS-B* if there exists a faithful normal state

$$\omega_{\text{sh}} \in (\mathcal{A}_{\text{sh}}^{\text{cand}})_*$$

such that:

1. $\omega_{\text{sh}} \circ \mathcal{E}_t^{(i)} = \omega_{\text{sh}}$ on $\mathcal{A}_{\text{sh}}^{\text{cand}}$ for every i and every $t \geq 0$;
2. ω_{sh} is compatible with the individual manifest states in the sense that
$$\omega_i|_{\mathcal{A}_{\text{sh}}^{\text{cand}}} = \omega_{\text{sh}} \qquad \forall i \in \{1, \ldots, N\}.$$

Remark 2.12 (JS-B is an agreement assumption, not a discovery). Faithfulness of each restriction follows from faithfulness of the independent ambient state ϖ_i, but equality of two restrictions does not. Clause (2) of JS-B is therefore the explicit substantive agreement input. If all ω_i were restrictions of the single reference state ω_0, clause (2) would be automatic and the discrepancy analysis would have no nonzero cases. The present datum deliberately excludes that shortcut.

Remark 2.13 (Strength of the JS-B stationarity clause). The stationarity requirement in Definition 2.11 is strong. It amounts to asking that the shared state lie in the common fixed-state sector of all source semigroups when restricted to $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$. The present paper does not claim that such a common stationary state is generic. A weaker variant would replace exact stationarity by long-time convergence or approximate stationarity on the shared sector; those relaxations are left open.

Definition 2.14 (JS-C: compatible persistent projections)**.** Let S_{multi} be a multi-source datum, and fix a candidate shared sector $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$. A pair of Nakajima–Zwanzig projections

$$\mathcal{P}^{(1)}, \mathcal{P}^{(2)} : \mathcal{A}_U \to \mathcal{A}_U$$

is called $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$-*compatible* if:

1. each $\mathcal{P}^{(i)}$ is a normal idempotent projection used in the Stage A output of the i-th source chain;
2. they commute,
$$\mathcal{P}^{(1)}\mathcal{P}^{(2)} = \mathcal{P}^{(2)}\mathcal{P}^{(1)} =: \mathcal{P}_{\mathrm{sh}};$$
3. $\mathcal{P}_{\mathrm{sh}}(\mathcal{A}_U)$ is a von Neumann subalgebra containing $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$, and $\mathcal{P}_{\mathrm{sh}}|_{\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}} = \mathrm{id}$;
4. the projected predual dynamics induced by the pair admit a common dual memory kernel
$$\mathcal{K}_{\mathrm{sh},*}(t, s)$$
on the shared projected sector, meaning that there exists a single kernel governing the shared projected dynamics on that sector; the definition does not require literal equality of the two individual source kernels before projection to the shared sector.

Remark 2.15 (Status of JS-C). Definition 2.14 is not imported from a standard open-systems textbook. It is a new named compatibility condition introduced for the present paper. The point is not to pretend that a model-independent notion of compatible Nakajima–Zwanzig projections is already standard. The point is to isolate exactly what later proofs need from such a notion. The background projection formalism is still the standard Nakajima–Zwanzig one [11, 12]; what is new here is only the multi-source compatibility layer. A weaker variant would require only that the long-time asymptotic behaviour of the two projected kernels agree on the shared sector, rather than demanding a common kernel at all times. The present paper uses the stronger form because it simplifies the Stage A construction; the weaker variant is left as an open relaxation.

Proposition 2.16 (Commuting expectations supply the algebraic part of JS-C)**.** *Let $E_i : \mathcal{A}_U \to \mathcal{M}_i$, $i = 1, 2$, be normal conditional expectations. If $E_1E_2 = E_2E_1$, then*

$$E_{12} := E_1E_2$$

is a normal conditional expectation onto $\mathcal{M}_1 \cap \mathcal{M}_2$. Consequently, if $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}} \subseteq \mathcal{M}_1 \cap \mathcal{M}_2$, clauses (1)–(3) of JS-C hold with $\mathcal{P}^{(i)} = E_i$. Clause (4), existence of a common projected memory kernel, remains an additional dynamical input.

Proof. Commutation and idempotence give $E_{12}^2 = E_1^2 E_2^2 = E_{12}$. Its range is contained in both $\mathcal{M}_1$ and $\mathcal{M}_2$, while every element of the intersection is fixed by both expectations. Thus $\operatorname{Ran} E_{12} = \mathcal{M}_1 \cap \mathcal{M}_2$. Normality, complete positivity, and unitality pass to the composition; the resulting norm-one positive projection is a conditional expectation. This is the commuting-expectation/commuting-square mechanism, not an automatic property of arbitrary Nakajima–Zwanzig projections [14, 15]. □

Proposition 2.17 (Finite-dimensional two-source witness)**.** *There is a finite Type I model with independent source states satisfying JS-A, JS-B, and JS-C and with $k_{\mathrm{sh}} = 1$. In the same model, the two manifest states agree on the shared sector but have nonzero discrepancy elsewhere in their overlap.*

Proof. Let

$$\mathcal{A}_U = M_2(S) \otimes M_2(P_1) \otimes M_2(P_2)$$

in its tracial standard form, and let Z_X be Pauli Z on factor X. Write

$$D_X(A) := \tfrac{1}{2}(A + Z_X A Z_X)$$

for the dephasing expectation on factor X, tensored with the identity on the other factors. Set

$$\mathcal{P}^{(1)} = D_S D_{P_1}, \qquad \mathcal{P}^{(2)} = D_S D_{P_2}.$$

These normal UCP conditional expectations commute, and $\mathcal{P}_{\mathrm{sh}} = D_S D_{P_1} D_{P_2}$ has the full diagonal algebra as its range. For $\lambda_i > 0$, define

$$\mathcal{E}_t^{(i)} = e^{-\lambda_i t}\,\mathrm{id} + (1 - e^{-\lambda_i t})\mathcal{P}^{(i)}.$$

Idempotence of $\mathcal{P}^{(i)}$ gives the semigroup law; each map is normal, UCP, and the two semigroups commute.

Take

$$\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}} = W^*(Z_S) \otimes \mathbf{1} \otimes \mathbf{1}, \qquad \mathcal{A}_c^{(1)} = \mathcal{A}_c^{(2)} = W^*(Z_S, Z_{P_1}, Z_{P_2}).$$

The reference tracial modular flow is trivial, and both semigroups fix the candidate shared sector pointwise, proving JS-A. Let $\rho(x) := \frac{1}{2}(\mathbf{1} + xZ)$, with $|x| < 1$, and choose the independent faithful ambient states

$$\varpi_1 = \rho(1/3) \otimes \rho(1/4) \otimes \rho(1/5), \qquad \varpi_2 = \rho(1/3) \otimes \rho(-1/4) \otimes \rho(-1/5).$$

Their manifest restrictions agree faithfully on the shared sector, with common state induced by $\rho(1/3)$, and that state is stationary there. This proves JS-B. Proposition 2.16 proves the algebraic clauses of JS-C. On the pointwise-fixed shared sector the projected generator vanishes, so the common projected kernel may be taken as $\mathcal{K}_{\mathrm{sh},*} = 0$, proving clause (4).

Finally choose foreground images

$$V_{\mathrm{fg}}^{(1)} = \operatorname{span}\{Z_S, Z_{P_1}\}, \qquad V_{\mathrm{fg}}^{(2)} = \operatorname{span}\{Z_S, Z_{P_2}\}.$$

Their intersection is $\operatorname{span}\{Z_S\}$, hence $k_{\mathrm{sh}} = 1$. Yet $\omega_1(Z_{P_1}) = 1/4$ and $\omega_2(Z_{P_1}) = -1/4$, so the discrepancy is nonzero on the larger overlap algebra. This witness verifies compatibility in FD/I; its zero shared kernel and tracial modular flow are not a Type III realization of the full Foundation programme. □

3 The Two-Source Shared Manifest Sector

We now specialize to $N = 2$. The paper's main theorem is entirely two-source.

3.1 Joint source interface

Let

$$S^{(2)}_{\text{multi}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_1, \varpi_2\}, \{\mathcal{E}^{(1)}_t, \mathcal{E}^{(2)}_t\})$$

be a two-source datum. Assume that each source separately satisfies the named Foundation I hypotheses required to produce a Stage A output, Stage B output, and manifest pair

$$(\mathcal{A}^{(i)}_c, \omega_i), \qquad i = 1, 2.$$

Lemma 3.1 (Joint Stage A output from compatible persistent projections)**.** *Let $S^{(2)}_{\text{multi}}$ be a two-source datum. Assume there exists a candidate shared sector $\mathcal{A}^{\text{cand}}_{\text{sh}} \subseteq \mathcal{A}^{(1)}_c \cap \mathcal{A}^{(2)}_c$ satisfying JS-C. Then the commuting product*

$$\mathcal{P}_{\text{sh}} := \mathcal{P}^{(1)}\mathcal{P}^{(2)}$$

defines a shared projected record object

$$\mathcal{O}^{\text{sh}}_A := (\mathcal{P}_{\text{sh}}, \omega_{\text{sh},t}, \mathcal{K}_{\text{sh},*}(t, s)),$$

where $\omega_{\text{sh},t}$ is the projected normal state on the shared sector and $\mathcal{K}_{\text{sh},*}(t, s)$ is the common dual memory kernel from Definition 2.14.

Proof. Commutativity of $\mathcal{P}^{(1)}$ and $\mathcal{P}^{(2)}$ implies that $\mathcal{P}_{\text{sh}}$ is again an idempotent projection. Because both $\mathcal{P}^{(1)}$ and $\mathcal{P}^{(2)}$ are normal and they commute, their composition is again normal. By JS-C, $\mathcal{P}_{\text{sh}}(\mathcal{A}_U)$ is a von Neumann subalgebra containing $\mathcal{A}^{\text{cand}}_{\text{sh}}$, and the common dual kernel $\mathcal{K}_{\text{sh},*}(t, s)$ is part of the named input. This is exactly the shared Stage A object required for later stages. □

3.2 Joint foreground overlap

For each source $i = 1, 2$, let $\mathfrak{F}^{(i)}$ denote the foreground frame produced by the conditional Foundation I finite-coordinate compression interface, with foreground realization

$$\iota_i : V_{\text{fg}}(\mathfrak{F}^{(i)}) \hookrightarrow \mathcal{A}^{\text{sa}}_U.$$

Definition 3.2 (Shared foreground overlap)**.** The *shared foreground overlap* is the operator-side intersection

$$V_{\text{fg}}(\mathfrak{F}_{\text{sh}}) := \iota_1\big(V_{\text{fg}}(\mathfrak{F}^{(1)})\big) \cap \iota_2\big(V_{\text{fg}}(\mathfrak{F}^{(2)})\big) \subseteq \mathcal{A}^{\text{sa}}_U.$$

Whenever this intersection is non-zero and finite-dimensional, we choose a basis

$$O^{\text{sh}}_1, \ldots, O^{\text{sh}}_{k_{\text{sh}}} \in V_{\text{fg}}(\mathfrak{F}_{\text{sh}})$$

and write

$$\mathcal{A}_{\mathfrak{F}_{\text{sh}}} := W^*(1, O^{\text{sh}}_1, \ldots, O^{\text{sh}}_{k_{\text{sh}}}).$$

Proposition 3.3 (Dimension bound for the shared foreground)**.** *Assume $\mathfrak{F}^{(1)}$ and $\mathfrak{F}^{(2)}$ have effective dimensions k_1^* and k_2^*, respectively, and that the shared foreground overlap of Definition 3.2 is finite dimensional. Then*

$$k_{\text{sh}} \leq \min\{k_1^*, k_2^*\}.$$

Proof. The shared foreground overlap is a subspace of each individual foreground image. Therefore its dimension cannot exceed the dimension of either image, hence cannot exceed the minimum of the two. □

Remark 3.4. Proposition 3.3 is only a bookkeeping result. It does not prove that the overlap is non-trivial. Non-triviality of the shared foreground remains a separate conditional input.

3.3 Shared stable hull

Write

$$\omega_0(\cdot) = \langle \Omega, \cdot\, \Omega \rangle$$

on the ambient algebra $\mathcal{A}_U$, and let $\sigma_s^{\omega_0,\text{tot}}$ be its ambient modular flow.

Definition 3.5 (Shared stable-carrier class)**.** Let $\mathfrak{F}_{\text{sh}}$ be a non-trivial shared foreground overlap. Define

$$\mathfrak{L}_{\text{sh}}^{\text{st}} := \left\{ \begin{array}{l} \mathcal{M} \subseteq \mathcal{A}_U : \mathcal{M} \text{ is a von Neumann subalgebra,} \\ \qquad \mathcal{A}_{\mathfrak{F}_{\text{sh}}} \subseteq \mathcal{M}, \\ \qquad \sigma_s^{\omega_0,\text{tot}}(\mathcal{M}) = \mathcal{M} \quad \forall s \in \mathbb{R}, \\ \qquad \mathcal{E}_t^{(i)}(\mathcal{M}) \subseteq \mathcal{M} \quad \forall t \geq 0,\ i = 1, 2 \end{array} \right\}.$$

Definition 3.6 (Shared stable hull)**.** Assume $\mathfrak{L}_{\text{sh}}^{\text{st}} \neq \varnothing$. The *shared stable hull* of the shared foreground is

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) := \bigcap_{\mathcal{M} \in \mathfrak{L}_{\text{sh}}^{\text{st}}} \mathcal{M}.$$

Proposition 3.7 (Minimal shared stable hull)**.** *Assume $\mathfrak{L}_{\text{sh}}^{\text{st}} \neq \varnothing$. Then*

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})$$

is a well-defined von Neumann subalgebra of $\mathcal{A}_U$ satisfying:

1. $\mathcal{A}_{\mathfrak{F}_{\text{sh}}} \subseteq \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})$;
2. $$\sigma_s^{\omega_0,\text{tot}}\big(\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})\big) = \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \qquad \forall s \in \mathbb{R};$$
3. $$\mathcal{E}_t^{(i)}\big(\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})\big) \subseteq \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \qquad \forall t \geq 0,\ i = 1, 2;$$
4. *it is minimal with these properties: if $\mathcal{N} \subseteq \mathcal{A}_U$ is any von Neumann subalgebra containing $\mathcal{A}_{\mathfrak{F}_{\text{sh}}}$ and stable under the ambient modular flow and both semigroups, then*

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \subseteq \mathcal{N}.$$

Proof. The argument is identical to the stable-hull construction in Foundation I [1], with the only change that one now imposes stability under both source semigroups. An arbitrary intersection of von Neumann subalgebras is again a von Neumann subalgebra: intersections of σ-weakly closed $*$-subalgebras remain σ-weakly closed $*$-subalgebras. Because every member of $\mathfrak{L}^{\mathrm{st}}_{\mathrm{sh}}$ contains $\mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}}$, so does the intersection. Stability under the ambient modular flow and both semigroups is preserved by intersection: if $A \in \bigcap_\alpha \mathcal{M}_\alpha$, then $\mathcal{E}_t^{(i)}(A) \in \mathcal{M}_\alpha$ for every α, t, and i, and similarly for the modular flow, hence the image again lies in $\bigcap_\alpha \mathcal{M}_\alpha$. Minimality is immediate from the definition as the intersection of all subalgebras with the required properties. □

3.4 Main theorem

Theorem 3.8 (Two-source shared admissible manifest sector)**.** *Let $S^{(2)}_{\mathrm{multi}}$ be a two-source datum. Assume:*

1. ***Individual source closure.*** *Each source separately satisfies the named Foundation I hypotheses needed to produce*

$$(\mathcal{A}_c^{(1)}, \omega_1), \qquad (\mathcal{A}_c^{(2)}, \omega_2),$$

   together with foreground frames $\mathfrak{F}^{(1)}$ and $\mathfrak{F}^{(2)}$.

2. ***Joint-stability package.*** *There exists a candidate shared sector*

$$\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}} \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}$$

   satisfying JS-A, JS-B, and JS-C.

3. ***Non-trivial shared foreground.*** *The shared foreground overlap $\mathfrak{F}_{\mathrm{sh}}$ is non-zero and admits a realization by exact shared pointer-aligned observables $O_1^{\mathrm{sh}}, \dots, O_{k_{\mathrm{sh}}}^{\mathrm{sh}}$ lying in $\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}}$.*

4. ***Admissibility input for Paper I.*** *The pair*

$$\left(\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}|_{\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})}\right)$$

   satisfies the admissibility conditions required by Paper I [2].

Then:

1. *the shared stable hull is well-defined and satisfies*

$$\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}} \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)};$$

2. *the restricted state*

$$\omega^{\mathrm{st}}_{\mathrm{sh}} := \omega_{\mathrm{sh}}|_{\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})}$$

   is faithful on $\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})$;

3. *the pair*

$$\left(\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}), \omega^{\mathrm{st}}_{\mathrm{sh}}\right)$$

   is a shared admissible manifest sector;

4. therefore the Paper I inseparability theorem applies to that pair.

Proof. By assumption (3), the exact shared pointer-aligned observables $O_1^{\mathrm{sh}}, \dots, O_{k_{\mathrm{sh}}}^{\mathrm{sh}}$ lie in $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$. Since $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$ is a von Neumann subalgebra by Definition 2.4, it contains the generated shared foreground algebra:

$$\mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}} \subseteq \mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}.$$

By assumption (2), the candidate shared sector satisfies JS-A, hence it is invariant under the ambient modular flow and under both source semigroups. Therefore

$$\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}} \in \mathfrak{L}_{\mathrm{sh}}^{\mathrm{st}}.$$

In particular, the class $\mathfrak{L}_{\mathrm{sh}}^{\mathrm{st}}$ is nonempty. Hence Definition 3.6 and Proposition 3.7 apply, so the shared stable hull is a well-defined von Neumann subalgebra of $\mathcal{A}_U$.

Because $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$ is a member of the shared stable-carrier class, minimality gives

$$\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}.$$

By assumption (2), the candidate shared sector is already contained in $\mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}$, so

$$\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

This proves part (1).

By JS-B, ω_{sh} is faithful on $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$. The restriction of a faithful normal state to a von Neumann subalgebra is again faithful. Therefore

$$\omega_{\mathrm{sh}}^{\mathrm{st}} = \omega_{\mathrm{sh}}|_{\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}})}$$

is faithful on the shared stable hull. This proves part (2).

Parts (1) and (2), together with the stability properties from Proposition 3.7, show that the pair

$$\left(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}}\right)$$

is a shared manifest sector in the sense intended by the present paper. Assumption (4) is the external admissibility input that upgrades it to a shared *admissible* manifest sector, so Paper I applies. This proves parts (3) and (4). □

Corollary 3.9 (Shared inseparability)**.** *Under the hypotheses of Theorem 3.8, geometry and modular flow on the shared sector*

$$\left(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}}\right)$$

are not independently deformable in the sense of Paper I. In particular, any geometric reading $\mathrm{Geom}(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}})$ *defined within the Paper I theorem domain is a* shared *geometric sector.*

Proof. This is exactly the conclusion of the Paper I theorem applied to the pair isolated by Theorem 3.8. □

Remark 3.10 (What the main theorem does *not* prove). Theorem 3.8 does not prove a general shared-world theorem. It proves something narrower: if a two-source datum already admits a candidate shared sector satisfying the named compatibility and admissibility inputs, then the overlap of the two source chains carries an honest theorem-level manifest sector to which the Paper I machinery may be applied. It also does not derive the candidate shared sector from multi-source entanglement or source dynamics alone. This is a foothold, not the full Bridge 2 closure.

Remark 3.11 (Where the theorem's new content lies). The genuine new content of Theorem 3.8 is the conditional inclusion chain

$$\text{shared foreground inclusion} \;+\; \text{JS-A} \Longrightarrow \mathcal{A}_{\text{sh}}^{\text{cand}} \in \mathfrak{L}_{\text{sh}}^{\text{st}} \Longrightarrow \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

The Paper I admissibility package is not derived here. It is a named external input specifying when the isolated shared stable hull lies in the theorem domain of Paper I.

4 Common/Private Decomposition

The main theorem isolates a shared admissible manifest sector. We now ask how each observer's full manifest algebra relates to that shared sector. In the Type III_1 setting, one should not treat this as a canonical factorization problem. The correct formulation is regime-dependent and uses a conditional expectation language.

Definition 4.1 (Shared/private regime assumption)**.** Let

$$\mathcal{A}_{\text{sh}} \subseteq \mathcal{A}_c^{(i)} \subseteq \mathcal{A}_U, \qquad i = 1, 2.$$

We say that the pair $(\mathcal{A}_c^{(i)}, \mathcal{A}_{\text{sh}})$ satisfies the *shared/private regime assumption* if for each i there exist:

1. a von Neumann subalgebra $\mathcal{A}_{\text{priv}}^{(i)} \subseteq \mathcal{A}_c^{(i)}$;
2. a faithful normal conditional expectation

   $$E_{\text{sh}}^{(i)} : \mathcal{A}_c^{(i)} \to \mathcal{A}_{\text{sh}}$$

   preserving the individual manifest state in the sense that

   $$\omega_i \circ E_{\text{sh}}^{(i)} = \omega_i;$$

3. a generating relation

   $$\mathcal{A}_c^{(i)} = W^*\big(\mathcal{A}_{\text{sh}} \cup \mathcal{A}_{\text{priv}}^{(i)}\big).$$

Remark 4.2. Definition 4.1 does not assert uniqueness of the private complement, and it does not assert a tensor-product or free-product factorization. It only says that, under a regime assumption, the full manifest algebra can be generated from the shared sector together with a private complement, with a faithful conditional expectation projecting back to the shared sector.

Remark 4.3 (Minimal package for common/private language). Definition 4.1 is the minimal package needed before common/private language can even be used honestly in the present paper. Without a private complement, a faithful conditional expectation onto the shared sector, and a generating relation for the full manifest algebra, the phrase "private part" remains interpretive rather than mathematical.

Proposition 4.4 (Unpacking JS-B and localizing overlap discrepancy)**.** *Assume the hypotheses of Theorem 3.8, and suppose the shared/private regime assumption holds with*

$$\mathcal{A}_{\text{sh}} := \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}).$$

Let

$$\mathcal{A}_{\text{ov}} := \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

Define the overlap discrepancy functional

$$\delta_{12} := \omega_1|_{\mathcal{A}_{\text{ov}}} - \omega_2|_{\mathcal{A}_{\text{ov}}} \in (\mathcal{A}_{\text{ov}})_*.$$

Then:

1. *the two restricted manifest states agree on the shared sector:*

$$\omega_1|_{\mathcal{A}_{\text{sh}}} = \omega_2|_{\mathcal{A}_{\text{sh}}} = \omega_{\text{sh}}^{\text{st}}|_{\mathcal{A}_{\text{sh}}};$$

2. *equivalently,*

$$\delta_{12}|_{\mathcal{A}_{\text{sh}}} = 0;$$

3. *if $A \in \mathcal{A}_{\text{ov}}$ satisfies $\delta_{12}(A) \neq 0$, then $A \notin \mathcal{A}_{\text{sh}}$;*

Proof. For $A \in \mathcal{A}_{\text{sh}}$, JS-B clause (2), restricted from the candidate shared sector to the shared stable hull, gives

$$\omega_1(A) = \omega_{\text{sh}}^{\text{st}}(A) = \omega_2(A).$$

Therefore

$$\delta_{12}(A) = 0 \qquad \forall A \in \mathcal{A}_{\text{sh}}.$$

This proves the first two claims. The third is only the contrapositive of the second; it is not a derived theorem that independent states must agree on the whole overlap. □

Remark 4.5 (Logical status of Proposition 4.4). The agreement statement is an unpacking of the explicit JS-B assumption. The proposition's useful content is bookkeeping: once independent states are part of the datum, it identifies where a nonzero overlap discrepancy is compatible with that assumption. Proposition 2.17 shows that this nonzero case is not vacuous.

Corollary 4.6 (Shared manifest content)**.** *Under the hypotheses of Proposition 4.4, every observable in the shared sector carries common manifest content. Observer-dependent discrepancy, insofar as it is jointly accessible on $\mathcal{A}_{\text{ov}}$, can only appear outside the shared sector.*

Proof. Proposition 4.4 shows that the two restricted manifest states agree on the shared sector and that any jointly accessible disagreement must be detected off that sector. That is the formal meaning of "shared manifest content" used in the present paper. □

Remark 4.7 (Shared/private reading of Proposition 4.4). Under the shared/private regime assumption, Proposition 4.4 may be read as saying that any overlap-level disagreement is detected only outside the common manifest sector, i.e. on the observer-dependent side of the decomposition language introduced in this section.

Remark 4.8 (What Section 4 does *not* prove). Section 4 does not prove existence of a canonical private factor, nor does it derive the shared/private regime assumption from the bare two-source datum. Its role is narrower: once a shared manifest sector has been isolated and a regime assumption is granted, it makes precise where observer agreement must hold and where observer-dependent discrepancy may still appear.

Remark 4.9 (No canonical factorization claim). Corollary 4.6 should not be read as a theorem that

$$\mathcal{A}_c^{(i)} \cong \mathcal{A}_{\text{sh}} \bar{\otimes} \mathcal{A}_{\text{priv}}^{(i)}$$

or any analogous canonical factorization holds. The paper makes no such claim. It uses only the weaker and safer statement that, under a regime assumption, a shared sector and a private complement jointly generate the full manifest algebra.

5 Conditional N-Source Extension

The two-source theorem is the paper's main result. The natural next step is a conditional N-source extension. This section records the generalization in the same honest style as Foundation I: whatever can be internalized immediately is internalized, and the genuinely new input is kept explicit.

Definition 5.1 (Joint N-source Stage A compatibility)**.** Let

$$S_{\text{multi}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_i\}_{i=1}^N, \{\mathcal{E}_t^{(i)}\}_{i=1}^N)$$

be a multi-source datum with individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$. A family of Stage A projections $\{\mathcal{P}^{(i)}\}_{i=1}^N$ is jointly compatible with $\mathcal{A}_{\text{sh}}^{\text{cand}}$ if:

1. the normal idempotents $\mathcal{P}^{(i)}$ commute pairwise;
2. the joint product
$$\mathcal{P}_{\text{sh}}^{(N)} := \prod_{i=1}^N \mathcal{P}^{(i)}$$
has von Neumann-algebra range containing $\mathcal{A}_{\text{sh}}^{\text{cand}}$ and fixes that candidate sector pointwise;
3. the jointly projected predual dynamics admits one common kernel $\mathcal{K}_{\text{sh},*,N}(t,s)$ on the joint range.

Lemma 5.2 (Joint N-source Stage A object)**.** *Under Definition 5.1, $\mathcal{P}_{\text{sh}}^{(N)}$ is a normal idempotent and, together with the stipulated common kernel, defines a joint Stage A object.*

Proof. Pairwise commutation makes the finite product independent of order and gives

$$(\mathcal{P}_{\text{sh}}^{(N)})^2 = \prod_{i=1}^N (\mathcal{P}^{(i)})^2 = \mathcal{P}_{\text{sh}}^{(N)}.$$

Finite composition preserves normality. The range and common-kernel requirements are explicit clauses of Definition 5.1, so the resulting tuple has the required Stage A data. Pairwise two-source theorems alone would not supply this one joint kernel or guarantee that all pairwise overlap choices define the same range. □

Theorem 5.3 (Conditional N-source shared manifest sector)**.** *Let S_{multi} be an N-source datum with individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$. Assume:*

1. *there exists a candidate shared sector*
$$\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \bigcap_{i=1}^N \mathcal{A}_c^{(i)}$$
satisfying JS-A and JS-B for all sources;
2. *the joint Stage A compatibility condition of Definition 5.1 holds, and a compatible family of shared foreground overlaps generates one common foreground algebra $\mathcal{A}_{\mathfrak{F}_{\text{sh}}}$;*

3. *the N-source stable-carrier class*

$$\mathfrak{L}^{\mathrm{st}}_{\mathrm{sh},N} := \left\{ \mathcal{M} \subseteq \mathcal{A}_U : \begin{array}{l} \mathcal{M} \text{ is a von Neumann subalgebra}, \\ \mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}} \subseteq \mathcal{M}, \\ \sigma_s^{\omega_0,\mathrm{tot}}(\mathcal{M}) = \mathcal{M},\ \forall s, \\ \mathcal{E}_t^{(i)}(\mathcal{M}) \subseteq \mathcal{M},\ \forall t, \forall i \end{array} \right\}$$

is nonempty and contains $\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}}$;

4. *the resulting N-source shared stable hull satisfies the admissibility hypotheses needed by Paper I.*

Then the N-source shared stable hull is a shared admissible manifest sector contained in $\bigcap_{i=1}^{N} \mathcal{A}_c^{(i)}$.

Proof. Lemma 5.2 supplies the joint Stage A object that pairwise compatibility alone does not provide. The remaining proof is the same minimality argument as in the two-source case, with the only change that one now intersects over all source labels rather than over two. The candidate shared sector is, by assumption, a member of the N-source stable-carrier class. Therefore the intersection defining the N-source shared stable hull is well-defined and lies inside the candidate sector. The remaining steps repeat the proof of Theorem 3.8. □

Remark 5.4 (What Theorem 5.3 does not contain). Theorem 5.3 does not provide:

1. a uniqueness or maximality theorem for the N-source shared sector;
2. a quantitative relationship between the number/strength of sources and geometric stability;
3. a proof that the joint Stage A and stable-carrier conditions of Definition 5.1 follow from the bare multi-source datum.

All three belong to later work.

6 Open Problems

The conditional inputs of Foundation II are not cosmetic. They isolate the genuinely unfinished parts of the programme.

Heuristically, the present two-source theorem is closer to a mean-field foothold than to a full correlated treatment: it isolates a shared effective manifest sector in a restricted regime, while genuinely cross-source structure remains the central open problem for any future multi-source closure.

1. **Model-independent joint stability.** The present paper names JS-A, JS-B, and JS-C. It does not derive them model-independently from a bare multi-source datum. The analogue of Foundation I's observer-emergence bottleneck therefore persists in multi-source form.
2. **Compatible Nakajima–Zwanzig projections.** The present notion of compatible persistent projections is a new definition, not a standard theorem. A general existence theory for such compatible projections remains open.

3. **Shared-foreground non-triviality.** The paper proves that if a non-trivial shared foreground overlap exists, then its dimension is bounded by the individual foreground dimensions. Proposition 2.17 gives one FD/I example with nonzero overlap, but no Type III or model-independent theorem forces the overlap to be nonzero.

4. **Quantitative consensus versus stability.** The paper deliberately avoids a bound of the form

$$\text{consensus strength} \Longrightarrow \text{geometric fluctuation bound}.$$

That is the separate Bridge 3 problem.

5. N**-source non-conditional closure.** The present N-source theorem is conditional. A truly non-conditional closure would require deriving a joint N-fold Stage A projection, common kernel, and common foreground from source-side dynamics.

6. **Uniqueness and maximality of the shared sector.** Even in the two-source case, the present theorem proves only existence of a minimal shared stable hull inside a candidate shared carrier. It does not prove that this sector is unique among all plausible shared manifest sectors, or maximal in any global sense.

7. **Relation to Paper II.** Paper II constrains higher-order geometry by multi-directional consistency [3]. The present paper does not yet prove how the shared manifest sector of Foundation II interacts with the Paper II small-ball remainder structure. That interface remains open.

8. **Type III and infinite-dimensional obstacles.** The common/private regime assumption is not automatic in the Type III setting. Conditional expectations, complements, and generating decompositions all become more delicate in infinite dimensions. In particular, existence of a faithful normal conditional expectation onto a candidate shared sector typically interacts with modular invariance in the sense of Takesaki and may fail outside suitable invariant regimes. Even when such expectations exist, complements and generating relations need not be unique, and non-factorial or center-bearing sectors may force a central decomposition before any shared/private language becomes well behaved.

7 Conditional Input Ledger

This section records the named inputs that remain conditional in the present paper.

1. **I0: Independent source states.** The faithful normal states ϖ_i are independent datum, and $\omega_i = \varpi_i|_{\mathcal{A}_c^{(i)}}$. They are not automatic restrictions of the reference vector state ω_0.

2. **Independent-state rebuild after retraction.** The former single-ambient-state reading is withdrawn because it made discrepancy vanish on the entire overlap. The current conditional theorem uses independent ϖ_i, and shared agreement enters explicitly as JS-B rather than as a derived fact.

3. **I1: Individual source closure.** Each source separately satisfies the named Foundation I hypotheses needed to produce an individual manifest pair $(\mathcal{A}_c^{(i)}, \omega_i)$.

4. **JS-A: Shared-sector invariance and semigroup compatibility.** The candidate shared sector is stable under all relevant semigroups and under the ambient modular flow, and the restricted semigroups commute there.

5. **JS-B: Shared-state existence and faithfulness.** A common faithful normal state exists on the candidate shared sector and is compatible with the source-specific manifest states.

6. **JS-C: Compatible persistent projections.** The pair of source-side Nakajima–Zwanzig projections commute and admit a common projected kernel on the shared sector. Commuting conditional expectations supply its algebraic clauses, but not the kernel clause.

7. **Finite compatibility witness.** Proposition 2.17 verifies JS-A/B/C and $k_{\text{sh}} = 1$ in a finite dephasing model with independent states. It is a model check, not a Type III emergence theorem.

8. **F1: Non-trivial shared foreground with exact pointer realization.** The overlap of the two source-side foreground realizations is non-zero, finite dimensional, and admits an exact pointer-aligned realization inside the candidate shared sector.

9. **A1: Admissibility input for Paper I.** The shared stable hull satisfies the admissibility package needed for the geometry–modular-flow inseparability theorem.

10. **SP1: Shared/private regime assumption.** Each individual manifest algebra admits a faithful conditional expectation onto the shared sector together with a private complement that generates the full algebra jointly with the shared sector.

11. **N1: joint N-source Stage A condition.** One jointly commuting product projection, a common range, and one common projected kernel are required. Pairwise two-source compatibility is insufficient by itself. No unused global consensus functional is retained.

A Imported Interfaces

This appendix records, in theorem-ready form, the external interfaces imported from Genesis, Foundation I, and the Consciousness Trilogy. They are not reproved here, but they are restated explicitly so that the present paper can be read without reconstructing its dependencies from the earlier series papers. What is *not* imported should also be kept clear: the multi-source notions of candidate shared sector, JS-A, JS-B, JS-C, shared stable hull, and shared/private regime are defined internally in Foundation II itself.

Proposition A.1 (Foundation I single-source closure package)**.** *Foundation I [1] isolates a conditional chain*

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where:

1. *Stage A supplies a projected record object*

$$\mathcal{O}_A = (\mathcal{P}, \omega_t, \mathcal{K}_*(t,s)),$$

where $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$ is a normal idempotent projection and $\mathcal{K}_*(t,s)$ is a dual non-Markovian memory kernel;

2. *Stage B supplies a foreground frame $\mathfrak{F}$ with a finite-dimensional foreground space $V_{\text{fg}}(\mathfrak{F})$ and an operator realization*

$$\iota : V_{\text{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\text{sa}};$$

3. *Stage C supplies a manifest pair*

$$(\mathcal{A}_c, \omega) = \big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}), \omega\big),$$

where $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_U$ is a minimal modular-decoherence stable hull, hence a von Neumann subalgebra carrying a faithful normal restricted state;

4. *in the exact fixed-point comparison regime imported from Foundation I, one has the inclusion*

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\mathrm{HAFF}}.$$

(not used in the present paper; recorded for later reference).

Proposition A.2 (Imported single-source interfaces used in Foundation II)**.** *Whenever Foundation II invokes the i-th source chain, the only external single-source objects it uses are:*

1. *a Stage A projection $\mathcal{P}^{(i)}$, projected normal states $\omega_t^{(i)}$, and dual memory kernel $\mathcal{K}_{*,i}(t,s)$;*
2. *a Stage B foreground frame $\mathfrak{F}^{(i)}$ with realized finite-dimensional foreground image*

$$\iota_i\big(V_{\mathrm{fg}}(\mathfrak{F}^{(i)})\big) \subseteq \mathcal{A}_U^{\mathrm{sa}};$$

3. *a Stage C manifest pair*

$$(\mathcal{A}_c^{(i)}, \omega_i) = \big(\Phi^{\mathrm{st}}_{\mathfrak{F}^{(i)}}(\mathfrak{F}^{(i)}), \omega_i\big).$$

No multi-source compatibility theorem is imported at this stage. Foundation II introduces the multi-source compatibility layer itself.

Proposition A.3 (Paper I manifest inseparability theorem)**.** *Paper I [2] proves that for an admissible manifest pair $(\mathcal{A}_c, \omega)$, geometry and modular flow are not independently deformable. In particular, once a pair $(\mathcal{A}_c, \omega)$ in the theorem domain is given, the geometry and modular sectors should be treated as two constrained readings of one underlying algebraic datum. Foundation II uses this result only through its admissibility interface: whenever a shared stable hull is shown, by named external input, to belong to the Paper I theorem domain, the inseparability conclusion becomes available.*

Proposition A.4 (Paper II as a consistency constraint)**.** *Paper II [3] supplies a conditional CHM small-ball stress-tensor extraction. Its stronger Einstein-tensor identification requires additional two-derivative and transversality hypotheses and is not imported here. Foundation II uses the paper only as boundary calibration, not as a formal input to the shared-sector existence proof.*

Proposition A.5 (T-DOME source-side ingredients)**.** *The T-DOME papers motivate three single-source interfaces used upstream of the present paper:*

1. *T-DOME I motivates the Stage A projected record process with kernel $\mathcal{K}_*(t,s)$, but does not prove that non-Markovian memory is necessary for persistence [7, 10];*
2. *T-DOME II motivates finite-coordinate foreground compression. Rank saturation and symmetry breaking require the explicit model assumptions stated in Foundation I; they are not universal necessity theorems [8];*

3. *T-DOME III: self-calibration enters through a Fisher-information prior carried by the foreground structure as a specified model input, but no direct theorem from T-DOME III is used in the operator-algebraic proofs below beyond this source-side calibration role [9].*

Proposition A.6 (HAFF ambient setting)**.** *The HAFF programme supplies the ambient structural setting used here:*

1. *geometry is read from accessible algebraic structure rather than supplied as a background container [5];*

2. *accessible algebras are selected by stability under the relevant dynamics rather than by subjective choice [6];*

3. *in the exact fixed-point regime used for Foundation I and imported here, the HAFF accessible algebra*

$$\mathcal{A}_c^{\mathrm{HAFF}} \subseteq \mathcal{A}_U$$

   is a von Neumann subalgebra stable under the relevant semigroup and the ambient modular flow;

4. *fixed-point and stability ideas for completely positive semigroups motivate the use of stable carrier classes and exact pointer sectors in restricted regimes [16, 17].*

Remark A.7 (What earlier papers do and do not supply). The earlier Genesis and Consciousness papers supply only single-source objects and single-source comparison statements of the kind listed above. They do *not* supply a multi-source theorem asserting that two projections commute, that two foreground frames overlap nontrivially, or that two source chains admit a common stationary state. Those are the precise new conditional interfaces named JS-A, JS-B, and JS-C in the main text.

Remark A.8 (Conditional expectations). The common/private regime assumption uses faithful normal conditional expectations in the sense of Takesaki [13]. No claim is made here that such expectations always exist onto the shared sector in the Type III_1 setting; their existence is part of the regime assumption itself.

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